Case Study 2 – Sales, Inventory & Operations Planning (SIOP) Reporting

Candidate Information

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Case Study: SIOP Reporting

Tool Used: Microsoft Power BI Desktop

1. Objective

The objective of this case study was to transform forecast snapshot data into a scalable reporting model that enables comparison of forecast values across different planning snapshots. The solution also integrates actual sales data to evaluate forecast performance using Forecast Accuracy and Forecast Bias KPIs.

The dashboard provides an executive-level view of Revenue, Non-Revenue, and CAPEX forecasts with interactive filtering and drill-down capabilities.

2. Dataset Description

The following datasets were used:

- Forecast Data
- Actual Data
- Calendar Table
- Entity Mapping
- Product Hierarchy

The Forecast dataset contains multiple forecast snapshots captured during different planning periods ("As Of Period"). Actual sales data is provided separately and merged with the forecast data for performance analysis.

3. Data Preparation

Data preparation was performed entirely in Power BI using Power Query.

The following transformations were applied:

- Imported all required Excel datasets.
- Removed unnecessary columns and corrected data types.
- Cleaned null and duplicate records.
- Created transformed forecast tables.
- Generated Resultant, Lag0, and Lag1 forecast values.
- Merged forecast data with actual sales data using Period, Country, and Item.
- Loaded Product Hierarchy, Calendar, and Entity Mapping tables.
- Established relationships between fact and dimension tables.

4. Data Model

A star schema data model was implemented.

The Forecast table acts as the central fact table and is connected with:

- Calendar Table
- Entity Mapping
- Product Hierarchy

This model enables efficient filtering and improves report performance.

5. DAX Measures

The following business measures were created:

- Total Revenue Forecast
- Total Non-Revenue Forecast
- Total CAPEX Forecast
- Revenue Actual

- Non-Revenue Actual
- CAPEX Actual
- Forecast Accuracy %
- Forecast Bias %

Forecast Accuracy was calculated using:

Forecast Accuracy = 1 − ABS(Forecast − Actual) / Actual

Forecast Bias was calculated using:

Forecast Bias = (Forecast − Actual) / Actual

6. Dashboard Features

The dashboard contains:

- Executive KPI Cards
- Revenue Forecast Analysis
- Non-Revenue Forecast Analysis
- CAPEX Forecast Analysis
- Country-wise Performance
- Trend Analysis
- Forecast vs Actual Comparison
- Dynamic Lag Selection (Resultant, Lag0, Lag1)
- Interactive slicers for Country, Period, Business Unit, Brand, and Forecast Snapshot

All visuals interact dynamically through Power BI relationships.

7. Key Insights

The dashboard enables business users to:

- Compare forecast versions across planning cycles.
- Measure forecast quality using Forecast Accuracy.
- Identify over-forecasting and under-forecasting through Forecast Bias.
- Analyze Revenue, Non-Revenue, and CAPEX forecasts by country and business hierarchy.
- Monitor forecast trends over multiple planning periods.

8. Challenges Faced

- Understanding forecast snapshot structure.
- Transforming multiple forecast versions into comparable lag columns.
- Maintaining scalable Power Query transformations.
- Creating a clean star schema model for efficient reporting.

9. Conclusion

The developed Power BI solution successfully transforms raw SIOP data into an interactive executive dashboard. The report provides accurate forecast comparison, integrates actual sales performance, and supports business decision-making through dynamic analysis and KPI tracking.

The solution is scalable and can accommodate future forecast snapshots without significant structural changes.